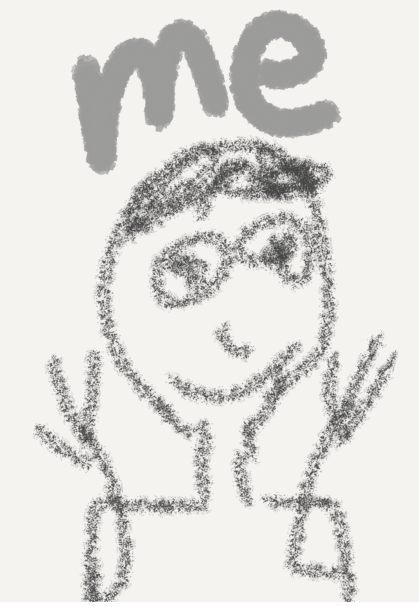

CHONG ZHANG

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Page: zita-ch.github.io

EDUCATION

2018 - present

B. Eng. (in progress)
Tsinghua University

Expected June, 2022.

My major is an interdiscipline major. It lies in the intersection of Computer Science, Mechanical Engineering, and Electrical Engineering.

My minor is offered by School of Software, with **two graduate courses** attended:

- Information Visualization and Visual Analytics
- Deep Learning

2015 - 2018

High School Diploma
Xishan Senior High School

PROFILE

Affiliation

Currently I am an **undergraduate** in **Tsinghua University**, China. I major in **Measurement, Control Technology and Instruments** (2018- present), with **Data Science and Technology Minor**.

Interests

My research interests lie primarily in robot control, machine Learning, and signal processing.

ACADEMIC & STANDARD TESTS

GPA and Rank

- Major: 3.88/4.00 (3/63)
- Minor: 3.94/4.00 (no rank)

English Proficiency

- TOEFL: 106 (R30/L26/S22/W28)

GRE SCORE

- GRE: 328 (V158/Q170) + 4.0

PUBLICATION

My GOOGLE SCHOLAR

- **Zhang, C.**, Yu, W., & Li, Z. (2021). Accessibility-Based Clustering for Efficient Learning of Robot Fall Recovery. arXiv preprint arXiv:2109.11191. [preprint] [video] (Submitted to RAL+ICRA2022)
- **Zhang, C.**, Hu, P., Zhang, R., Wu, Q., & Yang, A.. (2021). Modeling of Coupling Effects in Neural Networks for Ship Motion Prediction (Version 1). TechRxiv. https://doi.org/10.36227/techrxiv.14945544.v1 [preprint] (Submitted to IEEE Trans. on SMC)

PROFESSIONAL ACTIVITIES

Review

Reviewer for 2022 American Control Conference.

RESEARCH EXPERIENCE

Robocup Humanoid Jun. 2021 - present

Developing the gait with my teammates

For the gait of our humanoid soccer TH-MOS, we are trying to implement DRL methods for high-speed and robust locomotion, so as to achieve hierarchical control for actions such as dribbling and shooting.

Our group is currently **co-advised by Dr. Li Liu, Dr. Jiwen Zhang, and Dr. Chongjie Zhang**. Group members are from a wide range of different majors, including ME, EE, CS, and majors of liberal arts. We are all enthusiastic about robotics.

Robot Fall Recovery Jun. 2021 - present

Summer intern, advised by Dr. Zhibin (Alex) Li.

Cooperating with Alex and other researchers in the Advanced Intelligent Robotics (AIR) Lab, University of Edinburgh, I was engaged in studying the efficient learning methods for robot fall recovery. It feels good to work with them :D

For model-free DRL of robot fall recovery, it is an open research challenge in terms of data efficiency and robustness of learned policies. The technique of discovering initial state distributions with both high data efficiency and high robustness is of interest. In our work, we proposed and formulated novel algorithms to automatically discover a near-optimal set of initial states, based on un-supervised learing of clustering. We studied the data efficient exploration using these auto-discovered initial configurations of the robot, and achieved effective and efficient learning of DRL policies.

We submitted our work to RAL+ICRA2022. For details, see: [preprint] [video]. We are also trying to extend our previous research now.

Modelling and Prediction of Ship Motions Oct. 2019 - Jun. 2021

Person in charge, advised by Dr. Peida Hu.

This program started when I entered my sophomore year, and I managed to get the top-level funding from the National Training Programs of Innovation and Enterpreneurship for Undergraduate hosted by Ministry of Education in China.

Prediction is important for control. As part of the non-linearity in ship dynamics, coupling effects that affect motions of certain DOFs with other coupled DOFs are hard to model. In our work, we managed to apply factorization machines to model the coupling effects in the WD-RNN framework, thereby improving the prediction accuracy with low complexity. The proposed structure is interpretable because of its explicit modelling of the coupling effects.

We submitted our work to IEEE Trans. on SMC. For details, see: [preprint]

Topology Design for Multi-Chip Communication Jan. 2021 - Apr. 2021

Person in charge, advised by Dr. Guoqi Li.

This program was a follow-up of the previous research of which the results, the bio-inspired Tianjic chip, were published on Nature cover. To achieve the number of neurons in human brain (86B), millions of chips (156 cores * 256 neurons per chip) are required, and the communication efficiency of them is quite a challenge. Even supercomputers do not have so many CPU cores. We aimed to achieve a generic and near-optimal mapping method of cores and chips that could be suitable for any given mode of communication. We investigated many existing topology structures for communication, and proposed a heuristic structure for high communication efficiency.

HONORS AND AWARDS

Scholarship for Comprehensive Excellence Titled with December 9th. (2019). Highest level of scholarship in my major, 1 recipient out of 63 students.

Rank 6th in China Collegiate Computing Contest - Big Data Challenge 2020 (**national competition for college students**, with 2 teammates). The problem is to predict the ETA of the shipping orders.

And numerous prizes of school-level competitions...

SKILLS

Coding (mainly with Python/Matlab/C++), PCB design, embedded system development, mechanical design, Photoshop, office skills...
Also hobby: Pokémon Master

SOCIAL ACTIVITIES

Providing Equal Access to Engineering Training

I have been working in our department's **student association of technology** for three years, and currently I am the vice president. Our organization aims to develop students from diverse backgrounds who are interested in engineering to become experienced engineers. We provide funding, training and competitions for undergraduates, which are essential to improve their engineering skills. **Undergraduate students usually lack their own access to engineering training before entering labs for research, and it is hard for them to afford the cost, especially for those from poor families.**

During my first year here, I undertook some development tasks. During the second year, I worked as the leader of the competition committee. Currently, I am engaged in administrative routines.

COURSES

Math and Physics

Calculus, Linear Algebra, Probability and Statistics, Mathematical Modelling, Introduction to Complex Analysis, Physics, Theoretical Mechanics, Strength of Materials, Fundamentals of Engineering Thermodynamics...

Computer Science

Programming, Data Structure, The Fundamental of Computer: The Hardware/Software Interface, Introduction to Data Science, Information Retrieval Technology, Introduction to Big Data Systems, Fundamentals of Machine Learning, Intelligent Optimization Algorithms and Its Applications (attending), **Information Visualization and Visual Analytics** (graduate course), **Deep Learning** (graduate course)...

Electrical Engineering

Introduction to Intelligent Connected System, Electrical Engineering and Electronics, Design and Practice of Circuit System, Special Experiments of Electromechanical System...

Mechanical Engineering

Fundamentals of Machine Design, Application of Mechanical Graphics, Creative Design in Mechanics, Manufacturing Engineering Fundamentals...

Instrumentation and Control

General Chemistry, Introduction of Modern Biology, Fundamentals of Optical Engineering, Error Theory and Data Processing, Fundamental Technology for Instruments Design, Measurement and Test Technology, Precision Instruments Design, Micro-electromechanical System Design, Measurement Circuits and Systems, Fundamentals of Control Engineering, Computer Control Technology, MEMS Technique (attending), Sensors and Signals (attending), Numerical Control Technology (attending)...

*I prefer self-study and peer discussion rather than attending classes for new knowledge, so there are only a few graduate courses here. Btw, you can expect most of the courses listed here to be of grade 4.0.